Sai Kurelli

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Education

University of Texas at Austin Austin, TX

*Master of Science in **Computer Science** - 3.9 GPA* *2025*

*Bachelor of Science in **Computer Science**, Minor in **Business** - 3.9 GPA* *2025*

Technical Skills

Languages: Python, SQL, Scala, Java, C/C++, SwiftUI, React-Native with TypeScript
Tools: Apache Spark, Hadoop, Hive, Kafka, Docker, Kubernetes, AWS, PyTorch, GitHub

Experience

Software Engineer Aug 2026 – Present

Apple Inc.

- Creation of team **term dictionary** to improve on-boarding and consistency across workflows and documentation.

Software Engineer Jun 2025 – Aug 2026

Visa Inc.

- Built a **MCP server** in **Python** enabling natural language queries by leveraging **Claude** to translate user input into **SQL**, querying **Trino Fact Tables** for an accessible and secure data access interface.
- Redesigned **data pipelines** in **Scala** and **Java** to resolve unoptimized data flows across distributed databases, achieving a **40%** decrease in report generation time for an average of 2 million daily reports.
- Transformed fixed 1-second lag alerts into configurable data center-level thresholds between transactions, reducing false positives by **70%** for more accurate and meaningful monitoring.
- Accelerated **empty-report file write stage** from **2 hours** to **17 minutes** by migrating to **Java NIO** bulk batch writes, while preserving required header and footer rows.
- Replaced coarse pipeline delay checks, which caused **false-positive** and **false-negative** report-generation halts, with a monitoring system based on **Qlik** replication latency and per-report-type table monitors.

Software Engineer Intern May 2022 – Aug. 2022, May 2023 – Aug. 2023, May 2024 – Aug. 2024

Visa Inc.

- (S24) Designed a real-time analytics system for Hadoop Resource Utilization using **Apache Spark**, **Hive**, and **SQL** to optimize performance of a distributed queue with $\geq$ **4k Virtual Cores**, with insights visualized through **PowerBI** dashboards
- (S23) Built a **Spark** pipeline in **Scala** on Visa's **Hadoop File System** to extract, transform, and load **YARN** results for data analytics jobs across orgs, with compression and repartitioning for a **10x** reduction in file size
- (S22) Optimized 16 core **SQL** scripts with **JDBC** connections to **Spark-Thrift** servers for faster data processing and extraction workflows across large-scale datasets
- Designed systems with **JUnit** unit tests and integrated **Splunk** log monitoring across Agile sprints

Projects

MDP Modelling (GitHub Link): Designed a Programming Language with **WebPPL** in **Javascript** to model **Markov Decision Processes (MDP)** with probability embedded in the syntax. Applied the language to model a Blackjack game, implementing a value iteration algorithm to compute optimal policies and value functions.

HackMerced-VII Winner (Lablr): Created a **Vue.js** Chrome extension with **OpenCV** and **Python** to detect spatial pixel irregularities and assign tags on image distortions, with data extraction and integration pipelines on **AWS S3**.

Leadership

President (2023-2024), Vice President of Finances (2020-2023) Aug. 2020 – May 2024

Association for Computing Machinery *University of Texas*

- Designed and hosted community events to build an ACM community of more than 300 UT Students with an annual budget over $19,000
- Improved Academic Lesson Plans for CS 101s and pioneered new events like **AWS** and **Kafka** workshops